

AI powered speech-to-speech mock interview platform

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1. Introduction

1.1 Purpose

This Database Design Document (DDD) describes how data is organized, stored, and managed within the AI-powered speech-to-speech mock interview platform. It presents the logical structure of the database using an Entity-Relationship model, defines each table's attributes and constraints, and documents the design decisions behind the schema, including normalization, key design, and future scaling considerations.

1.2 Database Technology

SQLite is used for the MVP, chosen for its zero-configuration setup and file-based simplicity, which suits local, single-user development on a student's laptop at no cost. The schema is designed to be directly portable to PostgreSQL for a production deployment, since it avoids SQLite-specific features and uses standard relational constructs (foreign keys, unique constraints) supported identically by both systems. A relational database was chosen over a NoSQL/document store because the data is inherently structured and relational a session has a fixed set of questions, each question has at most one answer and the application benefits from referential integrity (e.g. an answer cannot exist without its question) that relational foreign key constraints enforce automatically.

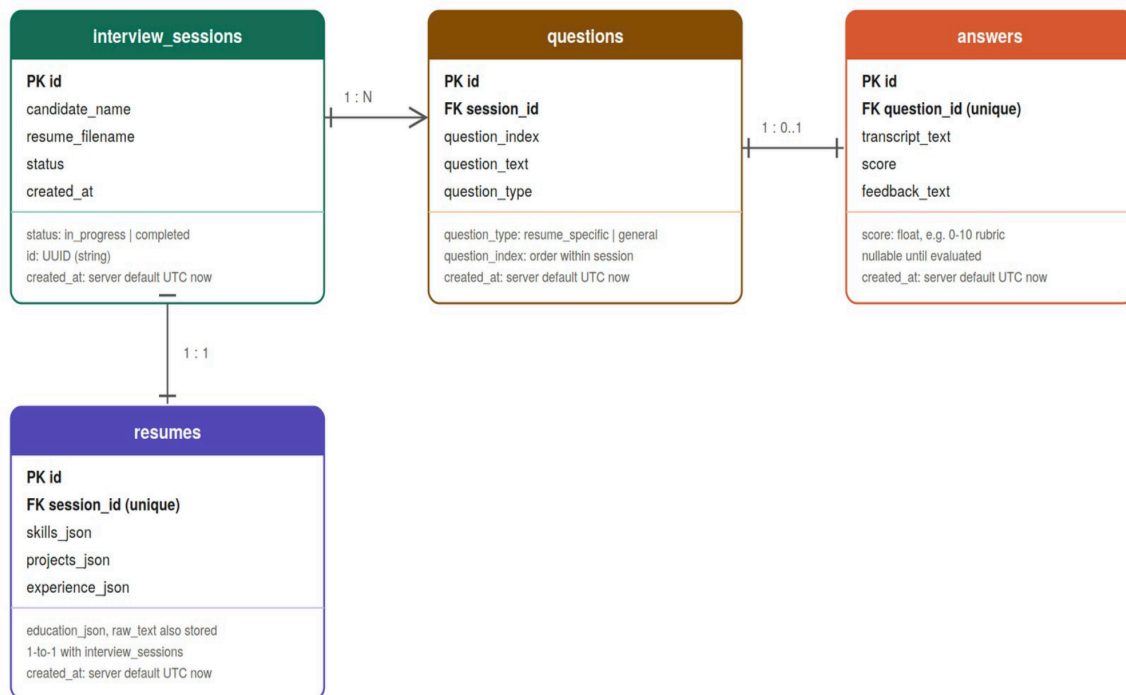
2. Conceptual Data Model

The system's data revolves around four core entities:

- Interview Session - represents one candidate's complete mock interview attempt, from resume upload to final report.
- Resume - the structured data extracted from the candidate's uploaded resume, tied one-to-one to a session.
- Question - a single interview question generated for a session, ordered by position.
- Answer - feedback. the candidate's transcribed spoken response to a question, along with its evaluation score and feedback.

A session has exactly one resume record and many questions; each question has at most one answer (none until the candidate responds). This mirrors the natural structure of the interview flow described in the System Architecture Document's sequence diagram.

3. Entity - Relationship Diagram



4. Database

4.1 Interview Sessions

Coloumn	Type	Constraints	Description
Id	String(UUID)	PK	Unique session identifier, generated on creation
candidate_name	String	Nullable	Candidates name, if provided
resume_filename	String	Nullable	Original filename of the uploaded resume
Status	Enum	Not null, default 'in_progress'	Session state: in_progress or completed
created_at	DateTime	Not null, default Now (UTC)	Timestamp the session was created

4.2 Resumes

Column	Type	Constraint	Description
Id	String (UUID)	PK	Unique resume record identifier
session_id	String (UUID)	FK -> interview sessions, id, unique, not null	Enforces the one to one relationship with a session
skills_json	Text (JSON)	Nullable	List of extracted technical skills
Experience_json	Text (JSON)	Nullable	List of project objects
education_json	Text (JSON)	Nullable	List of past roles/internships
raw_text	Text	Nullable	Cleaned raw extracted text
Created_at	DateTime	Not null, default now (UTC)	Timestamp the resume was parsed

4.3 Questions

Column	Type	Constraint	Description
Id	String (UUID)	PK	Unique Question identifier
session_id	String (UUID)	FK -> interview sessions, id, unique, not null	Session this question belongs to
question_index	Integer	Not Null	Position of the question within the session
question_text	Text	Not Null	The interview question shown/spoken to the candidate
question_type	String	Default 'general'	Resume specifier
created_at	DateTime	Not null, default now (UTC)	Timestamp the question was generated

4.4 Answers

Column	Type	Constraint	Description
Id	String (UUID)	PK	Unique answer identifier
question_id	String (UUID)	FK -> question. id, unique, not null	Enforces one answer per question
Transcript_text	Text	Nullable until submitted	Transcribed text of the candidates spoken answer
Score	Float	Nullable until evaluated	Rubric - based score
feedback_text	Text	Nullable until evaluated	Short qualitative feedback
created_at	DateTime	Not null, default now (UTC)	Time stamp the answer was submitted

5. Key Constraints & Summary

- All primary keys are UUID strings, generated in application code rather than relying on auto-increment integers this avoids exposing sequential IDs and allows IDs to be safely generated before a database write occurs.
- Foreign key constraints enforce that a question cannot exist without a valid session, an answer cannot exist without a valid question, and a resume cannot exist without a valid session.
- Unique constraints on resumes.session_id and answers.question_id enforce the 1:1 and 1:0..1 cardinalities respectively at the database level, not just in application logic.
- NOT NULL constraints are applied to every field required for the record to be meaningful (e.g. question_text, session status), while fields that are legitimately unknown until a later pipeline stage (score, feedback_text, transcript_text) are nullable.